HARDWARE DOCUMENT

VNT-002

First edition
2026-10-25

Vantage — Power board — Description and test

Applies to: Vantage power board, Rev A (Vantage_Power.kicad_sch, Rev A)

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Foreword

Team sevenUP prepares and maintains the hardware documents of the Vantage sensor node. The design files, the drawings and these documents are kept together in the team hardware repository, so that every hardware revision has a matching set of documents.

This document was prepared by Team sevenUP.

This first edition applies to Rev A of the power board schematic. The printed circuit board of the power board has not yet been laid out; fabrication and assembly instructions will be added in a later edition, when the layout is released. The changes between revisions are recorded in the repository change log (`CHANGELOG.md`).

Questions and proposed changes to this document should be raised as issues in the hardware repository.

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Introduction

The Vantage sensor node is the field unit of a flood-monitoring system. It is built from two boards connected by one cable: the sensor board (see VNT-001) and the power board described here.

The power board keeps the node running from a single 18650 lithium-ion cell charged by a small solar panel. It provides the 3.3 V supply of the ESP32-S3, a 5 V supply for the sensors that the ESP32-S3 switches on only while it measures, and two current monitors that report the solar and battery currents over I2C. The board is built mainly from ready-made modules, so that a faulty function can be repaired by replacing one module.

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1 Scope

This document describes the circuit of the Vantage power board, Rev A, its interfaces and configuration, and specifies the tests that a power board shall pass before it is connected to a sensor board.

It does not cover the sensor board (see VNT-001), the firmware, the solar panel mounting or the enclosure.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For each, the Rev A file in the hardware repository applies.

— `power-board/Vantage_Power.kicad_sch`, *Vantage — Power board — Schematic*

— `outputs/power-board/Vantage_Power_BOM.csv`, *Vantage — Power board — Bill of materials*

— VNT-001, *Vantage — Sensor board — Fabrication, assembly and test*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in VNT-001 and the following apply.

3.1
VBAT

internal supply rail of the power board, at battery voltage, from which the 3.3 V and 5 V supplies are made

3.2
maximum power point tracking

charger function that keeps the solar panel at the voltage at which it delivers the most power

3.3
battery management system

protection circuit that disconnects the cell on over-charge, over-discharge, over-current and short circuit

3.4
ideal diode

circuit that passes current in one direction with a very small voltage drop and blocks reverse current

3.5
sensor 5 V

switched 5 V supply for the sensors, delivered to J1 pin 4

4 Abbreviated terms

Abbreviation	Meaning
BMS	Battery management system
I2C	Inter-integrated circuit (two-wire serial bus)
MPPT	Maximum power point tracking
PTC	Positive temperature coefficient (resettable) fuse

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Abbreviation	Meaning
TVS	Transient voltage suppressor (diode)

5 Description

5.1 General

The power distribution is shown in Figure 1. The solar input and the battery both feed the VBAT rail. The 3.3 V supply and the sensor 5 V are made from VBAT.

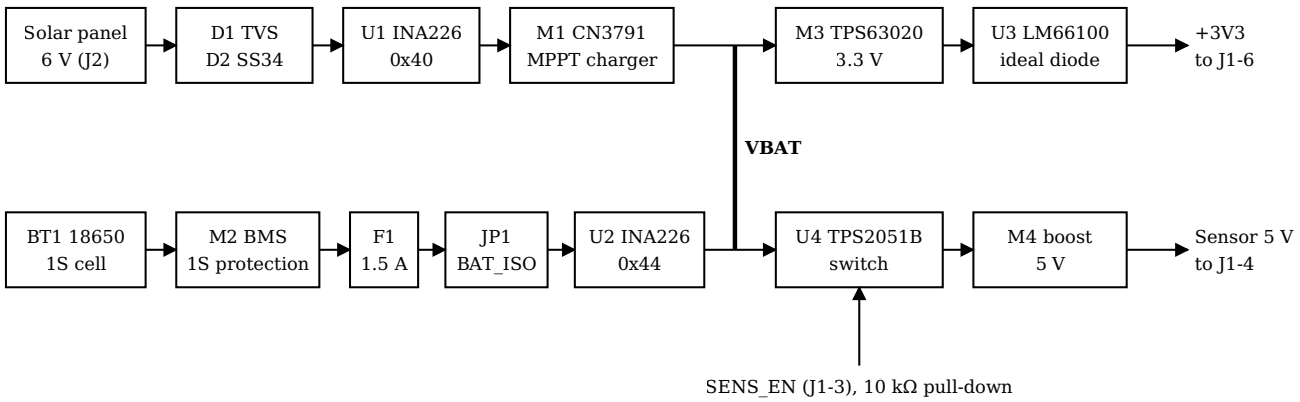


Figure 1 — Power distribution of the power board

5.2 Solar input

The solar panel connects to J2. D1 (SMAJ9.0CA) clamps voltage surges on the panel cable. D2 (SS34) blocks reverse current and protects against a reversed panel. U1, an INA226 module at I2C address 0x40, measures the panel voltage after D2 and the panel current. M1, a CN3791 MPPT charger module for a 6 V panel, charges the cell through the VBAT rail.

5.3 Battery

The cell BT1 plugs into a 2-way JST XH header. M2, a 1S protection module, disconnects the negative side of the cell on a fault; its B+ and P+ terminals are joined. F1 (PTC, 1.5 A) limits the battery current. JP1 (BAT_ISO) opens the battery path for storage, transport and current measurement. U2, an INA226 module at I2C address 0x44, measures the battery voltage and the battery current in both directions: positive while the cell discharges into VBAT, negative while M1 charges it.

C1 (220 µF) and C2 (100 nF) decouple VBAT. TP1 gives access to VBAT.

5.4 3.3 V supply

M3, a TPS63020 buck-boost module with 3.3 V output, keeps the 3.3 V supply steady over the whole cell voltage range. U3 (LM66100) is an ideal diode between M3 and the output; it stops the 3.3 V of the dev board, when the dev board is on USB, from flowing back into M3. Its enable input is tied to its output, which gives reverse blocking. C3 (220 µF) and C4 (100 nF) decouple the output. TP2 gives access to the 3.3 V supply, which goes to J1 pin 6, to the INA226 modules, to the I2C pull-ups and to J3.

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5.5 Sensor 5 V

U4 (TPS2051B) is a current-limited switch between VBAT and M4, a 5 V boost module. Its enable input is SENS_EN from J1 pin 3, with a 10 kΩ pull-down (R1), so the sensor 5 V is off unless the ESP32-S3 drives SENS_EN high. C5 (100 nF), C6 (10 µF), C7 (220 µF) and C8 (100 nF) decouple the switch and the boost module. TP3 gives access to the sensor 5 V, which goes to J1 pin 4.

5.6 I2C bus

The I2C bus runs from the sensor board through J1 pins 1 and 2 to U1 and U2. R2 and R3 (4.7 kΩ) are the only pull-up resistors on the bus. J3 brings the bus out for an extra I2C device or a test adapter.

6 Interfaces and configuration

6.1 Connectors and test points

Table 1 — Connectors and test points

Ref.	Function	Pins
J1	Sensor board connection, JST XH 6-way, vertical	1 I2C_SDA, 2 I2C_SCL, 3 SENS_EN, 4 sensor 5 V, 5 GND, 6 +3V3
J2	Solar panel, 2-way screw terminal 5.08 mm	1 panel +, 2 panel –
BT1	18650 cell lead, JST XH 2-way	1 cell +, 2 cell –
J3	I2C expansion, 4-way male header 2.54 mm	1 GND, 2 +3V3, 3 I2C_SDA, 4 I2C_SCL
TP1	VBAT	—
TP2	+3V3	—
TP3	Sensor 5 V	—
TP4	GND	—

The pin assignment of J1 shall match J1 of the sensor board (VNT-001, Annex C). The cable shall be wired pin 1 to pin 1 (VNT-001, 10.3).

6.2 Jumper JP1 (BAT_ISO)

JP1 shall be fitted for operation. JP1 may be opened to isolate the cell for storage or transport, or replaced by an ammeter to measure the battery current.

The board should not be run from the solar panel with JP1 open, because M1 would then work without a cell.

6.3 Module settings

The modules shall be set as follows before they are fitted:

a) U2 shall be set to I2C address 0x44 with the address pads of the module (A1 to VS, A0 to GND). U1 stays at the default address 0x40 (A1 and A0 to GND).

b) M1 shall be the version for a 6 V panel, or its maximum power point shall be set for the panel in use as given in the module documentation.

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c) The charge current of M1 shall not exceed the rating of the cell. It should not exceed 0.8 A, the measuring range of an INA226 module with a 0.1 Ω shunt (see Annex B).

d) M3 shall be a fixed 3.3 V output version. M4 shall have a 5 V output and accept an input from 3.0 V to 4.2 V.

7 Safety

WARNING — A lithium-ion cell can deliver very high current. Do not short its terminals or its lead. Connect the cell last and disconnect it first. Do not use or charge a damaged, swollen or deeply discharged cell.

WARNING — Check the polarity of the cell lead and of the panel cable before connecting them.

8 Prototype construction

Until the printed circuit board is released, a prototype may be built on stripboard or perforated board following the schematic. In that case:

— the battery and VBAT paths (BT1, M2, F1, JP1, U2, M1 output, M3 and U4 inputs) shall be wired with copper of at least 0.5 mm² cross-section;

— the ground connections of all modules shall return to one common ground point near C1;

— U3 and U4 are small surface-mount parts (SC-70 and SOT-23) and should be mounted on adapter boards;

— J1 shall be wired exactly as in Table 1.

9 Tests

9.1 General

The tests shall be made in the order of this clause, before the power board is connected to a sensor board. Equipment: a bench supply with an adjustable current limit, a digital multimeter, and an I2C controller (for example the sensor board connected through a test cable, or a USB I2C adapter on J3). The results should be recorded on the form in Annex C.

9.2 Visual and resistance checks

With no cell and no panel connected, check that all modules are fitted in the right orientation, and that the resistance from each of TP1, TP2 and TP3 to TP4 is above 1 kΩ once the capacitors have charged.

9.3 Supply tests without the cell

Open JP1. Set the bench supply to 3.7 V with a current limit of 0.3 A and connect it between TP1 (+) and TP4 (−). The results shall be as given in Table 2.

Table 2 — Supply tests

No.	Test	Pass condition
S-1	Supply current, SENS_EN open	Below 50 mA
S-2	Voltage TP2 to TP4	3.20 V to 3.40 V
S-3	Voltage TP3 to TP4, SENS_EN (J1-3) open	Below 0.3 V

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No.	Test	Pass condition
S-4	Voltage TP3 to TP4, SENS_EN (J1-3) connected to +3V3 (J3-2)	4.75 V to 5.25 V
S-5	Voltage TP2 to TP4 with the bench supply set to 3.0 V, then 4.2 V	3.20 V to 3.40 V at both settings

9.4 I2C tests

With the bench supply still connected, connect the I2C controller and scan the bus. The results shall be as given in Table 3.

Table 3 — I2C tests

No.	Test	Pass condition
I-1	Bus scan	Devices found at 0x40 and 0x44, and no others
I-2	U1 bus voltage and current, no panel	Close to 0 V and 0 mA

9.5 Tests with the cell and the panel

Disconnect the bench supply. Check that the cell voltage is between 3.0 V and 4.2 V and that the polarity of its lead is correct, plug it into BT1 and fit JP1. Then connect the panel in daylight, or a bench supply set to 6.0 V with a current limit of 0.5 A, to J2. The results shall be as given in Table 4.

Table 4 — Tests with the cell and the panel

No.	Test	Pass condition
B-1	U2 bus voltage compared with the voltage TP1 to TP4	Within ± 0.05 V
B-2	U2 current, no panel, SENS_EN open	Positive (discharge), below 50 mA
B-3	U1 bus voltage with the panel connected	Panel voltage less 0.3 V to 0.5 V (drop of D2)
B-4	U2 current with the panel connected and M1 charging	Negative (charging)

9.6 Test with the sensor board

Connect the sensor board as specified in VNT-001, 10.3 and carry out the field-mode tests of VNT-001, 11.6.

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Annex A
(normative)
Bill of materials

`Vantage_Power_BOM.csv` holds the same list in machine-readable form.

Table A.1 — Bill of materials

Item	Qty	Reference	Value	Package / part
1	1	BT1	18650 1S	JST B2B-XH-A, to the 18650 cell lead
2	2	C1, C7	220 µF 16 V	Radial electrolytic, 6.3 mm, 2.5 mm pitch
3	4	C2, C4, C5, C8	100 nF	1206
4	1	C3	220 µF 10 V	Radial electrolytic, 6.3 mm, 2.5 mm pitch
5	2	C6, C9	10 µF	1206
6	1	D1	SMAJ9.0CA	SMA, bidirectional TVS
7	1	D2	SS34	SMA, 3 A Schottky
8	1	F1	PTC 1.5 A	1812, 1.5 A hold
9	1	J1	SENSOR	JST B6B-XH-A, XH 6-way, vertical
10	1	J2	SOLAR 6V	2-way screw terminal, 5.08 mm
11	1	J3	I2C	1 × 4 male header, 2.54 mm
12	1	JP1	BAT_ISO	1 × 2 male header, 2.54 mm, with jumper cap
13	1	M1	CN3791 MPPT 6V	CN3791 MPPT charger module, 6 V panel
14	1	M2	BMS 1S	1S lithium-ion protection module
15	1	M3	TPS63020 3V3	TPS63020 buck-boost module, 3.3 V output
16	1	M4	BOOST 5V	5 V boost module
17	1	R1	10 kΩ	1206
18	2	R2, R3	4.7 kΩ	1206
19	4	TP1 to TP4	VBAT, 3V3, 5V, GND	Test pad
20	2	U1, U2	INA226	INA226 module, addresses 0x40 (U1) and 0x44 (U2)
21	1	U3	LM66100	Ideal diode, SC-70-6
22	1	U4	TPS2051B	Power switch, SOT-23-5

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Annex B
(informative)
Current monitors

Common INA226 modules carry a 0.1 Ω shunt resistor, marked R100. The INA226 measures up to 81.92 mV across the shunt, so the measuring range is about ± 0.8 A. Table B.1 gives the settings that suit this shunt.

Table B.1 — INA226 settings for a 0.1 Ω shunt

Item	U1 (solar)	U2 (battery)
I2C address	0x40	0x44
Shunt resistance	0.1 Ω	0.1 Ω
Current resolution (current LSB)	0.1 mA	0.1 mA
Calibration register (0.00512 / (current LSB × shunt))	512	512
Bus voltage measured at	Panel side, after D2	Battery side, after JP1
Sign of the current	Positive from the panel	Positive on discharge, negative on charge

NOTE If a module carries another shunt value, recalculate the calibration register with the formula in the table and check the measuring range against the charge current of M1.

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Annex C
(informative)
Test record

Board serial number	Date	Tested by

Table C.1 — Test record

Test	Clause	Measured value	Pass / fail
Visual and resistance checks	9.2		
S-1	9.3		
S-2	9.3		
S-3	9.3		
S-4	9.3		
S-5	9.3		
I-1	9.4		
I-2	9.4		
B-1	9.5		
B-2	9.5		
B-3	9.5		
B-4	9.5		
Test with the sensor board	9.6		
Result: accepted / rejected			

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Bibliography

a) Texas Instruments, *INA226 High-Side or Low-Side Measurement, Bi-Directional Current and Power Monitor with I2C Compatible Interface — Datasheet*

b) Texas Instruments, *TPS63020 High Efficiency Single Inductor Buck-Boost Converter — Datasheet*

c) Texas Instruments, *LM66100 5.5-V, 1.5-A Ideal Diode — Datasheet*

d) Texas Instruments, *TPS2051B Current-Limited Power-Distribution Switch — Datasheet*

e) Consonance Electronics, *CN3791 Solar Lithium-ion Battery Charger with MPPT — Datasheet*

f) JST, *XH Connector — Catalogue*

g) ISO/IEC Directives, Part 2, *Principles and rules for the structure and drafting of ISO and IEC documents* (layout model of this document)

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